

1. (a) Note that

$$\begin{pmatrix} x \\ y \end{pmatrix} \in N(T) \iff T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \iff \begin{cases} 3x - 2y = 0 \\ x = 0 \\ x + 2y = 0 \end{cases} \iff \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

So $N(T) = \{0\}$. Moreover, the empty set $\emptyset$ is the basis for $N(T)$:

- $\text{span}(\emptyset) = \{0\}$.
- As we cannot find a finite number of distinct vectors $v_1, \dots, v_k \in \emptyset$ so that

$$a_1 v_1 + \dots + a_k v_k = 0$$

for some $a_j \in \mathbb{R}$, $\emptyset$ is linearly independent.

(b) Yes, it is because $T\left(\begin{pmatrix} 4 \\ 3 \end{pmatrix}\right) = \begin{pmatrix} 6 \\ 4 \\ 10 \end{pmatrix}$.

2. (a) By (1), $S = \text{span}(\{\mathbf{e}_1, \mathbf{e}_2\}) \subset N(T)$. So

$$\begin{aligned} \text{rank}(T) &= 4 - \dim(N(T)) \quad (\text{by the dimension formula}) \\ &\leq 4 - \dim(S) = 4 - 2 = 2. \end{aligned}$$

- (b) For any $a \in \mathbb{R}$, let $T_a : \mathbb{R}^4 \rightarrow \mathbb{R}^5$ be defined by

$$T_a\left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}\right) = \begin{pmatrix} ax_4 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Clearly, T_a is a linear transformation satisfying (1) and $T_a \neq T_b$ for $a \neq b \in \mathbb{R}$. So the set $\{T_a : a \in \mathbb{R}\}$ contains infinitely many linear transformations satisfying (1).

3. For any $x, y \in V$, if $T(x) = T(y)$, then

$$x = T(T(x)) - T(x) = T(T(y)) - T(y) = y.$$

So T is injective.

For any $y \in V$, we have

$$\begin{aligned} y &= T(T(y)) - T(y) \\ &= T(T(y) - y) \quad (\text{by linearity}) \\ &= T(x) \quad \text{by taking } x = T(y) - y \in V. \end{aligned}$$

So T is surjective.

4. (a) For any $x, y \in \mathbb{R}^2$ and any $c \in \mathbb{R}$,

$$\begin{aligned} \bullet \tilde{T}(x+y) &= \begin{pmatrix} T(2(x+y)) \\ T(x+y) - (x+y) \end{pmatrix} = \begin{pmatrix} T(2x+2y) \\ T(x)+T(y)-x-y \end{pmatrix} = \begin{pmatrix} T(2x)+T(2y) \\ T(x)-x+T(y)-y \end{pmatrix} \\ &= \begin{pmatrix} T(2x) \\ T(x)-x \end{pmatrix} + \begin{pmatrix} T(2y) \\ T(y)-y \end{pmatrix} = \tilde{T}(x) + \tilde{T}(y). \\ \bullet \tilde{T}(cx) &= \begin{pmatrix} T(2cx) \\ T(cx) - cx \end{pmatrix} = \begin{pmatrix} cT(2x) \\ cT(x) - cx \end{pmatrix} = c \begin{pmatrix} T(2x) \\ T(x) - x \end{pmatrix} = c\tilde{T}(x). \end{aligned}$$

So T is a linear transformation.

- (b) (i) (⊃) : It follows from $\tilde{T}(0_{2 \times 1}) = 0_{4 \times 1}$.
 (⊂) : For any $x \in \mathbb{R}^2$,

$$\begin{aligned} x \in N(\tilde{T}) &\implies \begin{pmatrix} 2T(x) \\ T(x) - x \end{pmatrix} = \begin{pmatrix} T(2x) \\ T(x) - x \end{pmatrix} = \tilde{T}(x) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ &\implies 2T(x) = T(x) - x = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \end{aligned}$$

(ii) For any $x \in \mathbb{R}^2$,

$$\begin{aligned} x \in N(\tilde{T} \circ T) &\iff \tilde{T} \circ T(x) = \tilde{T}(T(x)) = 0_{4 \times 1} \\ &\iff T(x) \in N(\tilde{T}) \xrightarrow{(b)(i)} \{0_{2 \times 1}\} \iff T(x) = 0_{2 \times 1} \iff x \in N(T). \end{aligned}$$

- (c) (i) Proof: If T is bijective, then $N(T) = \{0_{2 \times 1}\}$. By (b)(ii),

$$N(\tilde{T} \circ T) = N(T) = \{0_{2 \times 1}\}.$$

So $\tilde{T} \circ T$ is injective.

- (ii) Proof: If $\tilde{T} \circ T$ is injective, then $N(\tilde{T} \circ T) = \{0_{2 \times 1}\}$. By (b)(ii),

$$N(T) = N(\tilde{T} \circ T) = \{0_{2 \times 1}\}.$$

Thus, T is injective, and hence bijective as it is a linear transformation between two vector spaces with the same dimension.